

UPGRADE OF FRACTURE MECHANICS BY SPRESS-SPRAIN RELATIONS: LIMITING DAMAGE FIELD CURVATURE

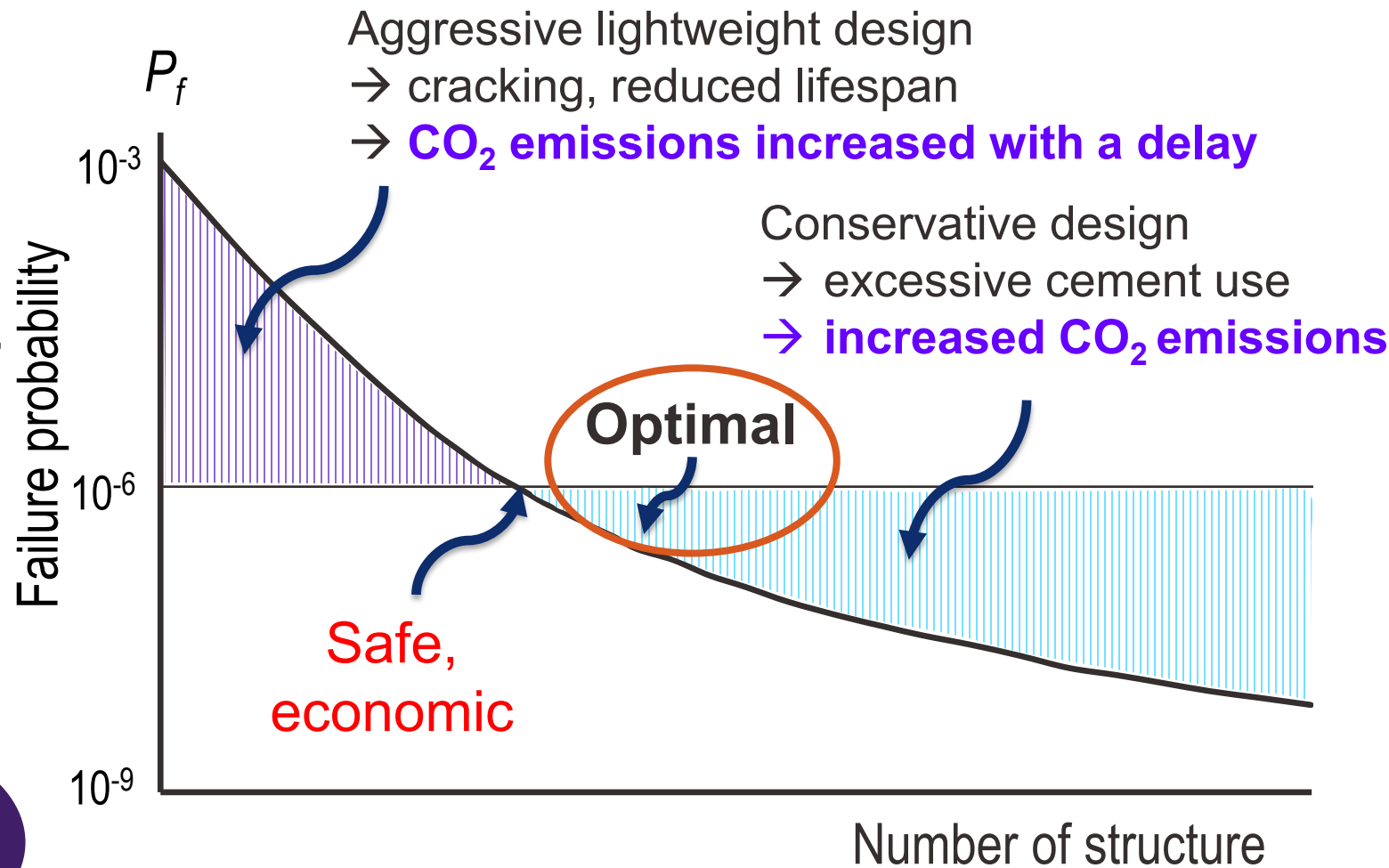
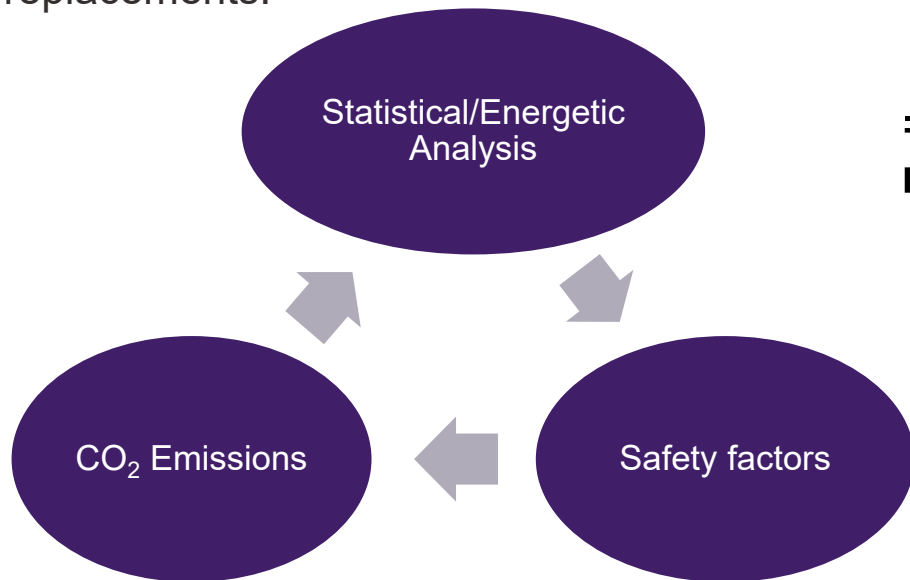
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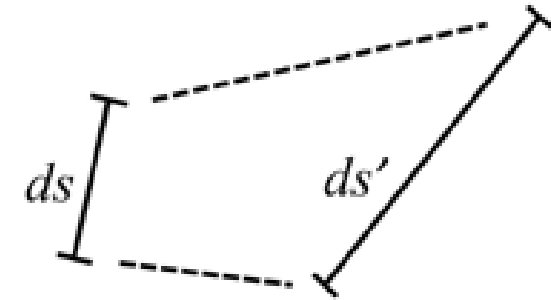
Safety Factors in Concrete: Impact on CO₂ Emissions

- **Safety Balancing CO₂:** Safety factors in concrete structures significantly impact CO₂ emissions from cement production.
- **Excess Safety, More Concrete:** High safety factors lead to unnecessary concrete use, increasing CO₂ emissions.
- **Inadequate Safety, Reduced Durability:** Insufficient safety factors result in more cracks, impairing durability and necessitating early replacements.



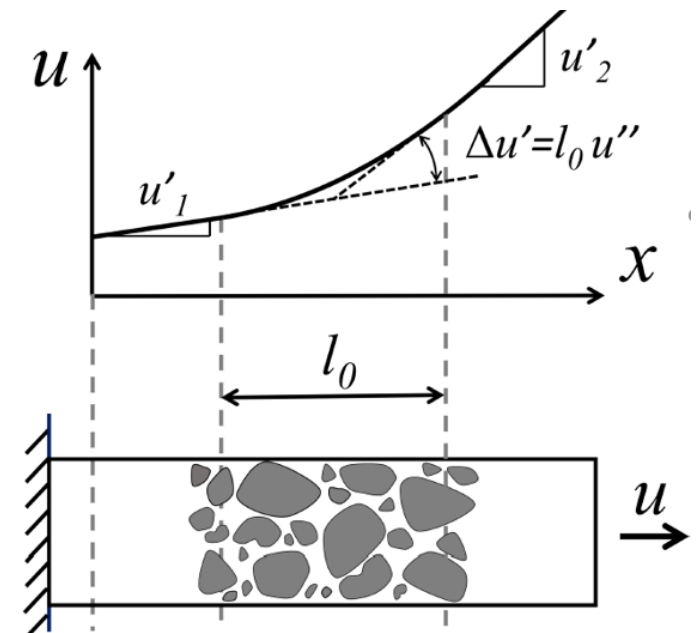
Homogenization with Sprain Energy

- If free energy density Φ depends on a change of **line segment length** only, it is a function of **strain tensor only**.



Traditional method

- Φ also depends on a **change of displacement gradient** $u_{i,j}$, or the **second displacement gradient** $u_{i,jk}$ called the **sprain**, which includes **rotation gradient**
- The change of displacement gradient $u_{i,j}$ must be distributed over a finite **material length** l_0 characterizing heterogeneity

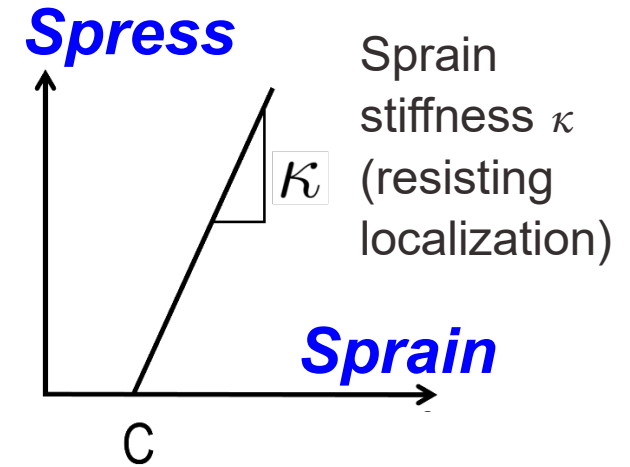


Sprain energy method

Total (Strain+Sprain) Helmholtz Free Energy

$$\bar{\Psi}(\epsilon, l_0 \eta) = \underbrace{\Psi(\epsilon)}_{\text{strain energy}} + \underbrace{\Phi(l_0 \eta)}_{\text{sprain energy}} = \text{total energy (density)}$$

sprain tensor



sprain stiffness

sprain threshold

For 1D $\Phi(x) = \frac{\kappa}{2} \langle l_0 |u''(x)| - C \rangle^2$

$$\Phi(x_r) = \frac{\kappa}{2} \left\langle l_0 \frac{|u_{r-1} - 2u_r + u_{r+1}|}{h^2} - C \right\rangle^2$$

Second gradient (or curvature) of displacement

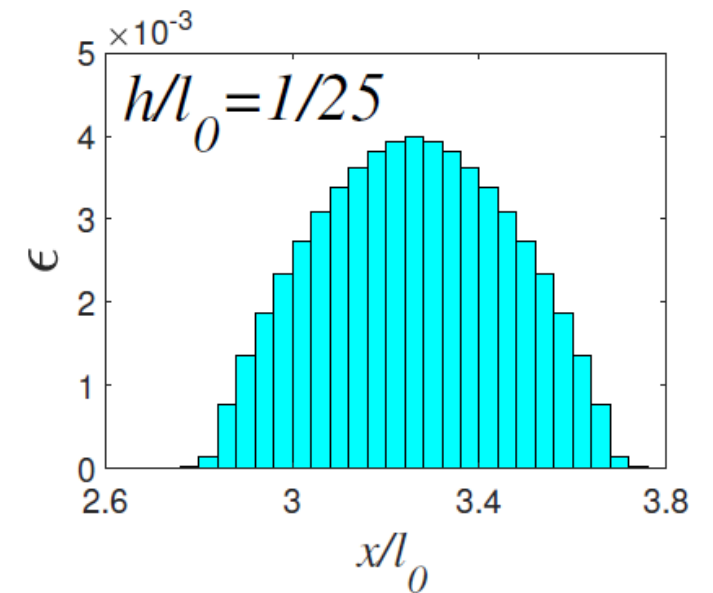
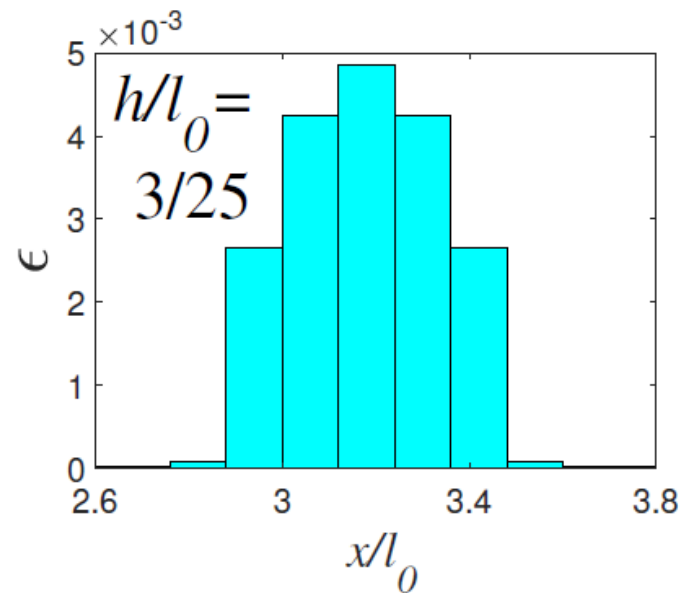
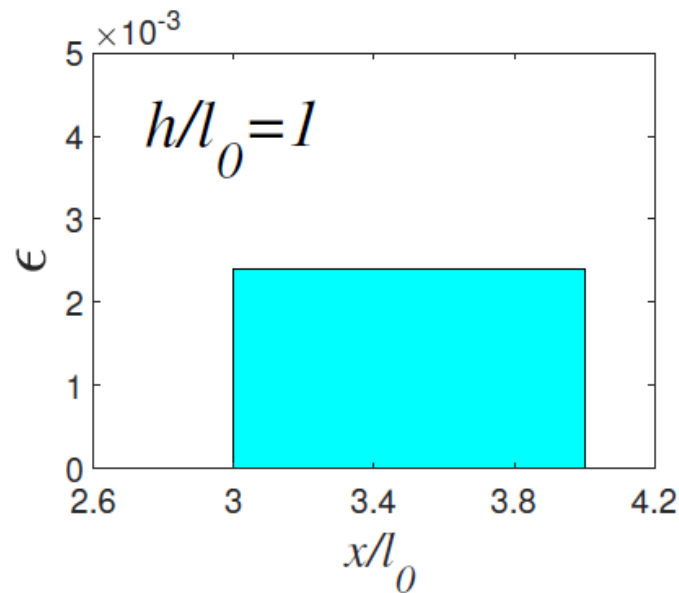
$$\eta_{ijk} = u_{i,jk} = \epsilon_{ij,k} + \omega_{ij,k}$$

= strain gradient + material rotation gradient

Smooth Crack Band Model (sCBM)

Convergence to smoothly distributed damage (1D bar in tension)

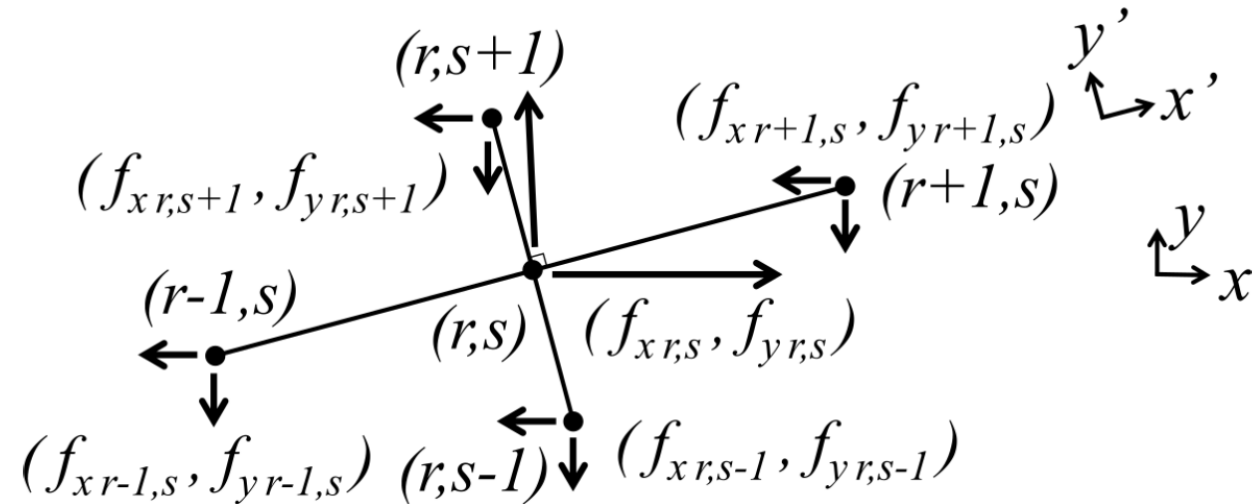
l_0 fixed



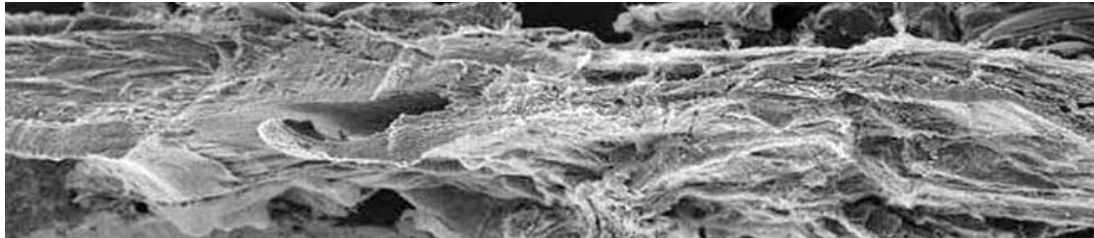
Nodal Sprain Forces in 2D

$$f_{x_{r,s}} = -h_x h_y t \frac{\partial \Phi_{r,s}}{\partial u_{x_{r,s}}}, \quad f_{x_{r-1,s}} = -h_x h_y t \frac{\partial \Phi_{r,s}}{\partial u_{x_{r-1,s}}}, \quad f_{x_{r+1,s}} = -h_x h_y t \frac{\partial \Phi_{r,s}}{\partial u_{x_{r+1,s}}}$$

$$f_{y_{r,s}} = -h_x h_y t \frac{\partial \Phi_{r,s}}{\partial u_{y_{r,s}}}, \quad f_{y_{r-1,s}} = -h_x h_y t \frac{\partial \Phi_{r,s}}{\partial u_{y_{r-1,s}}}, \quad f_{y_{r+1,s}} = -h_x h_y t \frac{\partial \Phi_{r,s}}{\partial u_{y_{r+1,s}}}$$



Sprain? Sprain!

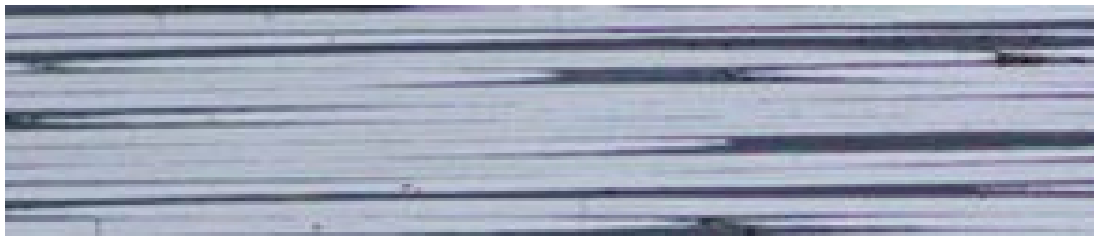


Nanoscale microscopic image of **ligament sprain in medicine** – overstretch with distributed local breaks (damage) over a finite length, but no full break*



Tensile microcracking in sprain of **concrete**

(<https://civillearners.com/2022/01/11/microcracking-of-concrete/>),



Distributed sprain damage in **carbon composites**

(from researchgate.net)

Total Helmholtz Free Energy Density

$$\bar{\Psi}(\boldsymbol{\varepsilon}, \boldsymbol{\xi}) = \underbrace{\Psi(\boldsymbol{\varepsilon})}_{\text{Strain energy}} + \underbrace{\Phi(\boldsymbol{\xi})}_{\text{Sprain energy}}$$

$$\xi_{ijk} = l_0 u_{i,jk}, \quad \eta_{ijk} = \varepsilon_{ij,k}, \quad u_{i,jk} = \eta_{ijk} + \omega_{ij,k}$$



Require high order element **How to implement it in C^0 element?**

Introduce intermediate variables

$$\boldsymbol{\zeta} = \nabla \mathbf{u} \quad \boldsymbol{\xi} = l_0 \nabla \boldsymbol{\zeta}$$

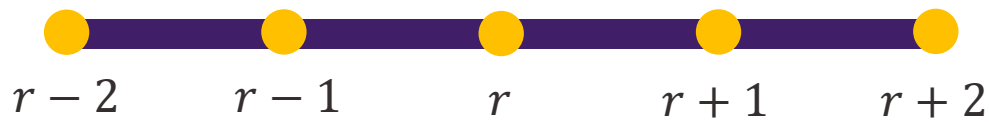
$$\Pi(\mathbf{u}, \boldsymbol{\zeta}, \boldsymbol{\lambda}) = W_{in} - W_{ex} \quad \text{where}$$

$$W_{in} = \int_V \Psi[\boldsymbol{\varepsilon}(\mathbf{u})] dV + \int_V \Phi[\boldsymbol{\xi}(\boldsymbol{\zeta})] dV - \int_V \boldsymbol{\lambda}(\nabla \mathbf{u} - \boldsymbol{\zeta}) dV$$

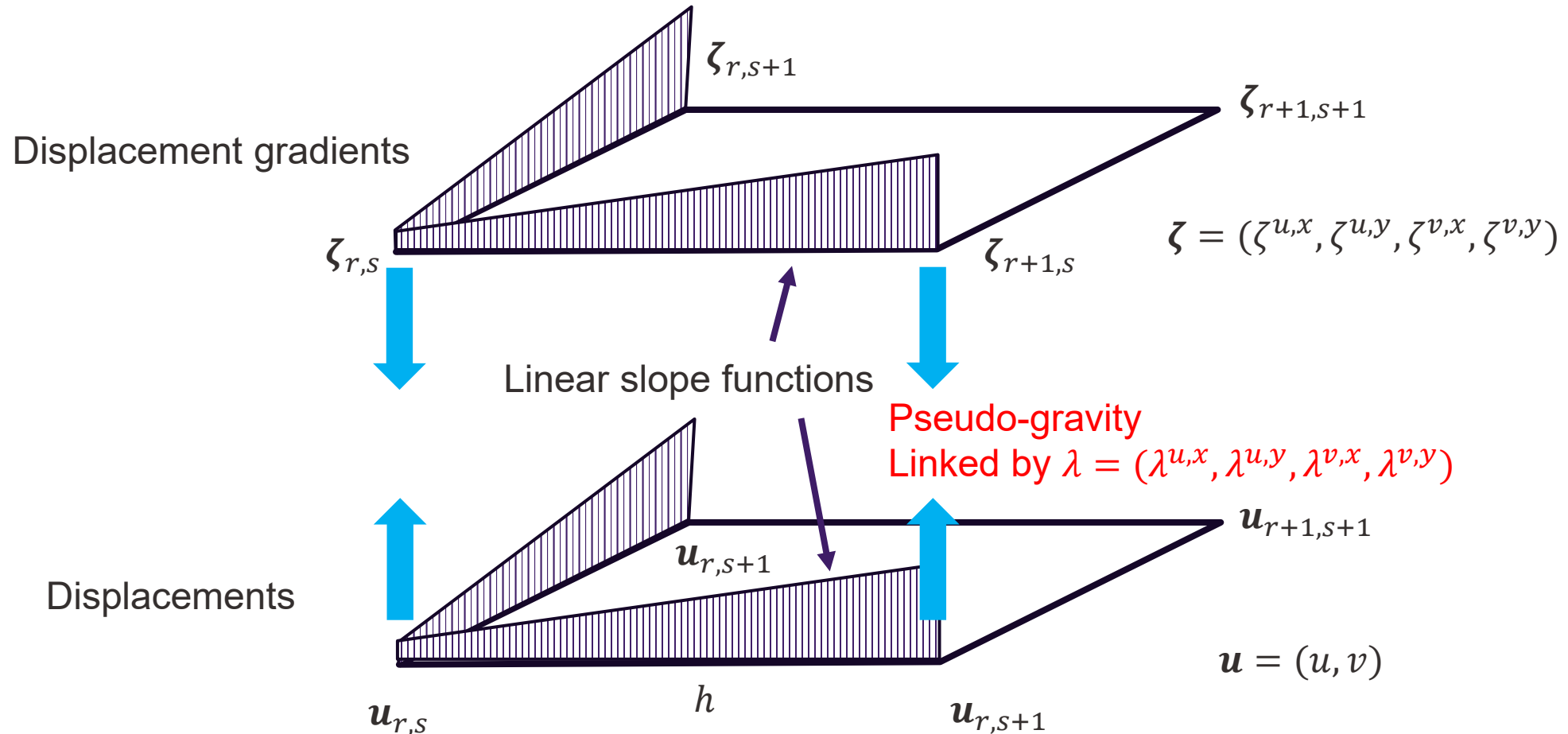
Constrained and coupled by Lagrange multiplier

$$W_{ex} = \int_V \mathbf{f}(\mathbf{x}) \cdot \mathbf{u} dV$$

Explanation of Lagrange multiplier λ



$$\frac{1}{h}(u_{r+1,s} - u_{r,s}) \approx \text{average}(\zeta_{r,s}, \zeta_{r+1,s}) \text{ via Lagrange multiplier } \lambda$$



Variational Analysis

Weak Form

$$\begin{aligned}\frac{\partial \Pi(u_i, \zeta_{ij}, \lambda_{ij})}{\partial u_i} \delta u_i &= \int_A \{ \sigma_{ij} \delta \varepsilon_{ij} - f_i \delta u_i - \lambda_{ij} \delta u_{i,j} \} dV \\ &= \int_V \{ -\sigma_{ij,j} - f_i + \lambda_{ij,j} \} \delta u_i dV + [\dots]_V = 0\end{aligned}$$

$$\begin{aligned}\frac{\partial \Pi(u_i, \zeta_{ij}, \lambda_{ij})}{\partial \zeta_{ij}} \delta \zeta_{ij} &= \int_V \{ s_{ijk} \delta \xi_{ijk} + \lambda_{ij} \delta \zeta_{ij} \} dV \\ &= \int_V \{ -l_0 s_{ijk,k} + \lambda_{ij} \} \delta \zeta_{ij} dV + [\dots]_V = 0\end{aligned}$$

$$\frac{\partial \Pi(u_i, \zeta_{ij}, \lambda_{ij})}{\partial \lambda_{ij}} \delta \lambda_{ij} = \int_V - \{ u_{i,j} - \zeta_{ij} \} \delta \lambda_{ij} dV = 0$$

Strong Form

$$\sigma_{ij,j} + f_i - \lambda_{ij,j} = 0$$



$$l_0 s_{ijk,k} - \lambda_{ij} = 0$$

$$u_{i,j} - \zeta_{ij} = 0$$

ABAQUS UEL Implementation

$$\bar{\mathbf{u}}_j = [u_j, v_j, \zeta_j^{u,x}, \zeta_j^{u,y}, \zeta_j^{v,x}, \zeta_j^{v,y}]^T \quad \text{where } j = 1, 2, 3, 4$$

$$\tilde{\boldsymbol{\lambda}}^T = [\lambda_1^{u,x}, \lambda_1^{u,y}, \lambda_1^{v,x}, \lambda_1^{v,y}, \lambda_2^{u,x}, \lambda_2^{u,y}, \lambda_2^{v,x}, \lambda_2^{v,y}, \lambda_3^{u,x}, \lambda_3^{u,y}, \lambda_3^{v,x}, \lambda_3^{v,y}, \lambda_4^{u,x}, \lambda_4^{u,y}, \lambda_4^{v,x}, \lambda_4^{v,y}]$$

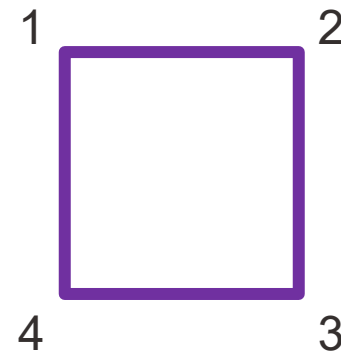
$$\mathbf{f}_{int}^u = \frac{\partial \Pi}{\partial \tilde{\mathbf{u}}^T} = b \left\{ \int_A \mathbf{B}_1^T \boldsymbol{\sigma} dA - \int_A \mathbf{B}_3^T \boldsymbol{\lambda} dA - \int_A \mathbf{N}^T(x, y) \mathbf{f}(x, y) dA \right\}$$

$$\mathbf{f}_{int}^\zeta = \frac{\partial \Pi}{\partial \tilde{\boldsymbol{\zeta}}^T} = b \left\{ \int_A \mathbf{B}_2^T \mathbf{s} dx + \int_A \mathbf{N}_\zeta^T(x, y) \boldsymbol{\lambda} dA \right\}$$

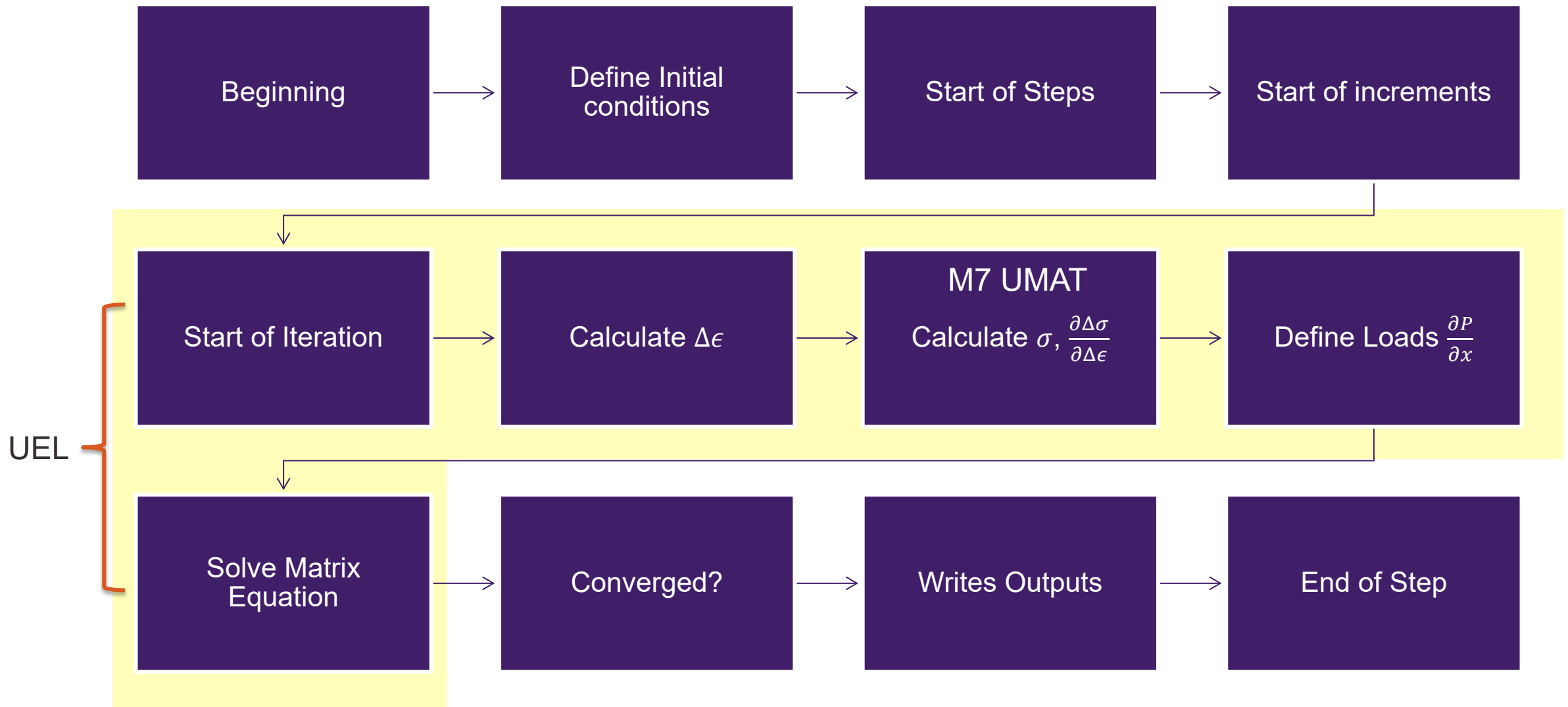
$$\mathbf{f}_{int}^\lambda = \frac{\partial \Pi}{\partial \boldsymbol{\lambda}^T} = b \int_A \{ \mathbf{B}_3 \tilde{\mathbf{u}} - \mathbf{N}_\zeta(x, y) \tilde{\boldsymbol{\zeta}} \} dA$$

Nodal DOFs:

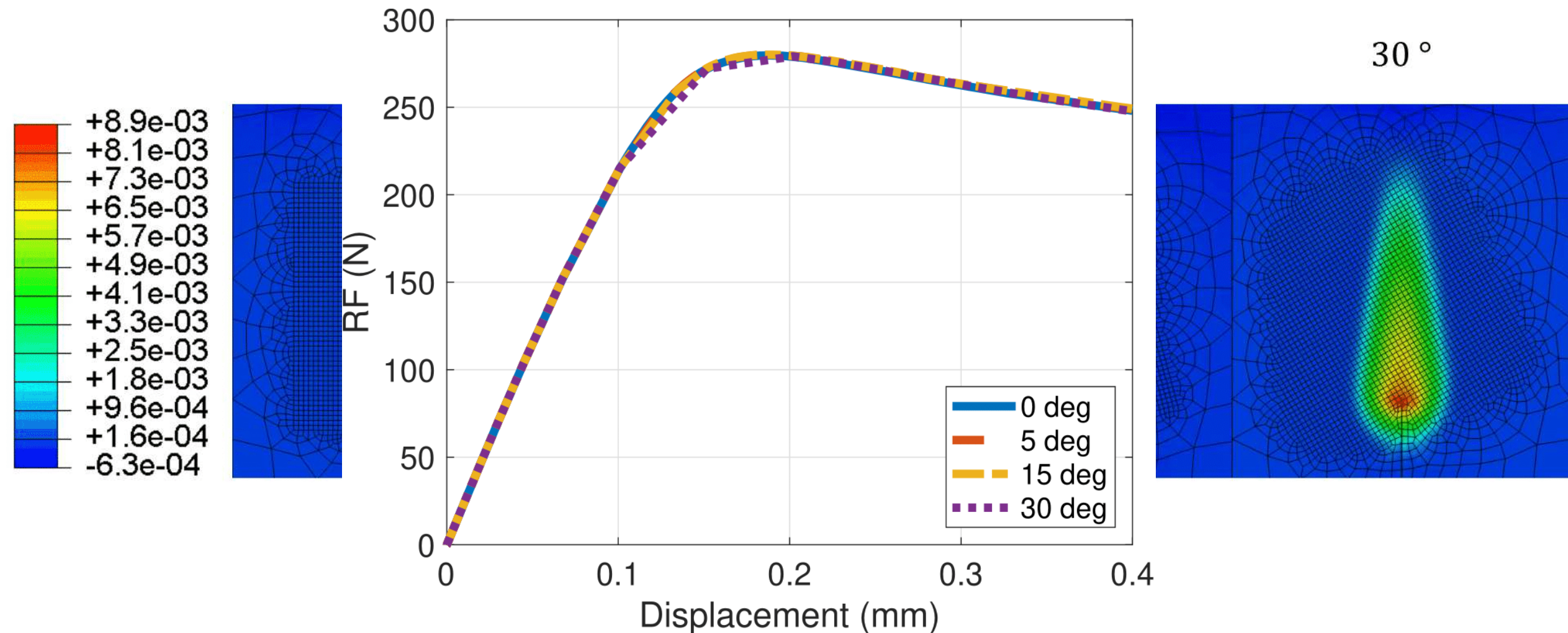
$u_1, v_1,$
 $\zeta_1^{u,x}, \zeta_1^{u,y}, \zeta_1^{v,x}, \zeta_1^{v,y},$
 $\lambda_1^{u,x}, \lambda_1^{u,y}, \lambda_1^{v,x}, \lambda_1^{v,y}$



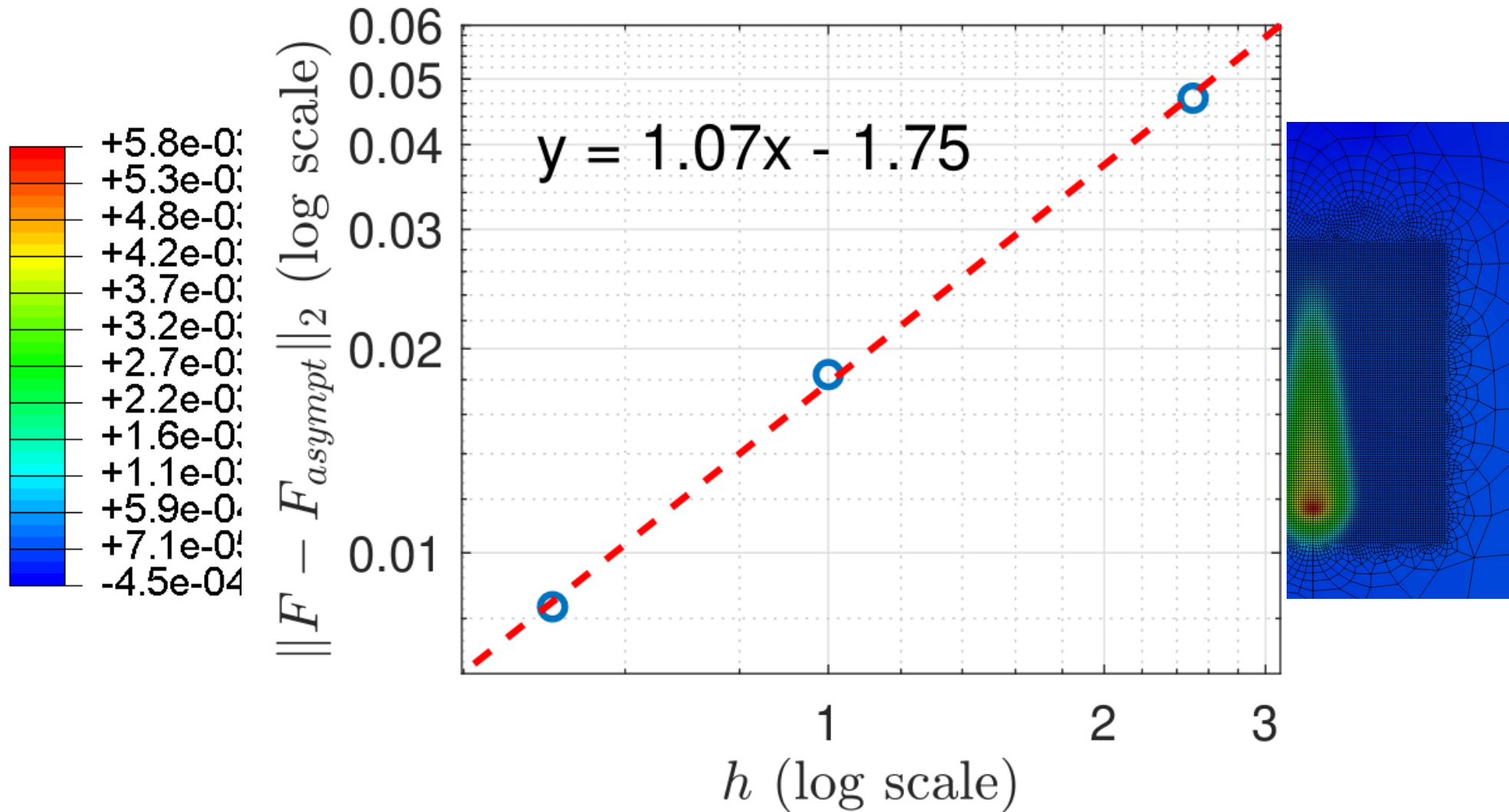
ABAQUS UEL Implementation



Three-point Bend Simulations with Different Mesh Inclinations using the sICBM and M7

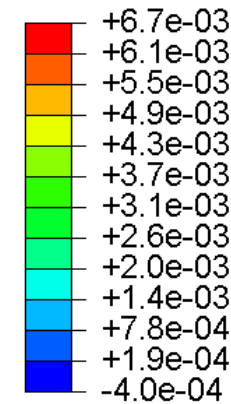
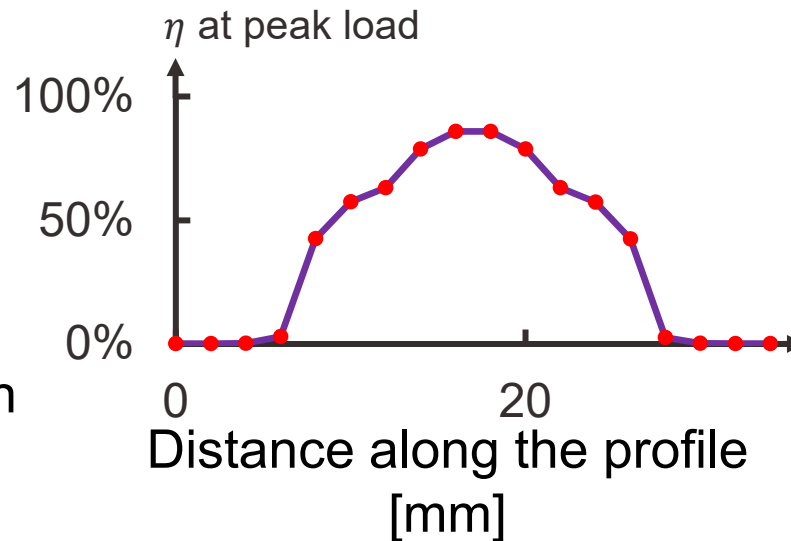
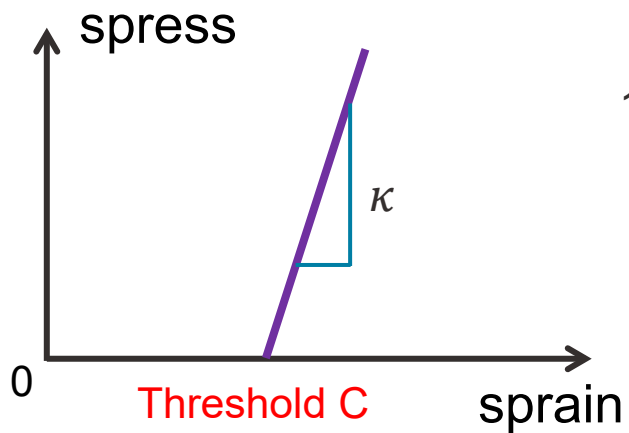
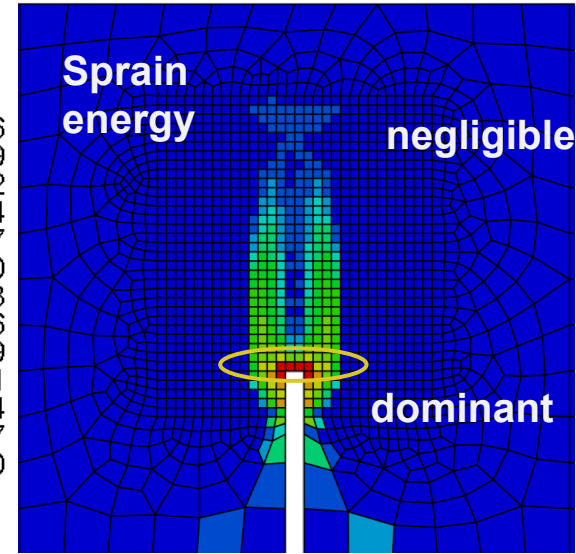
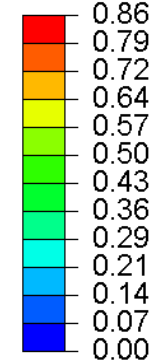
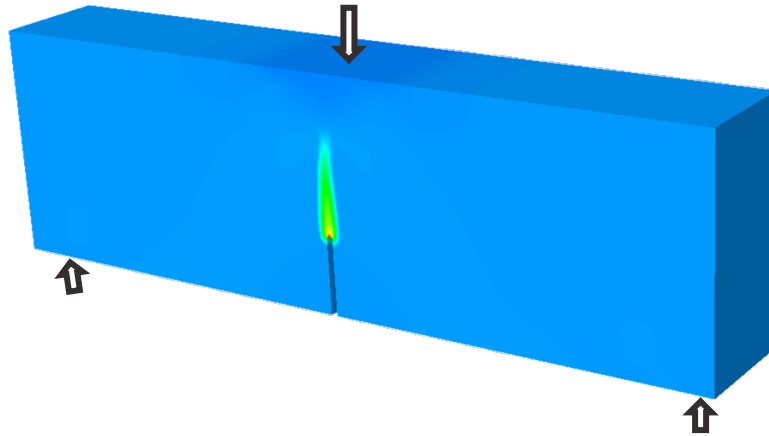


Convergence Check of sICBM

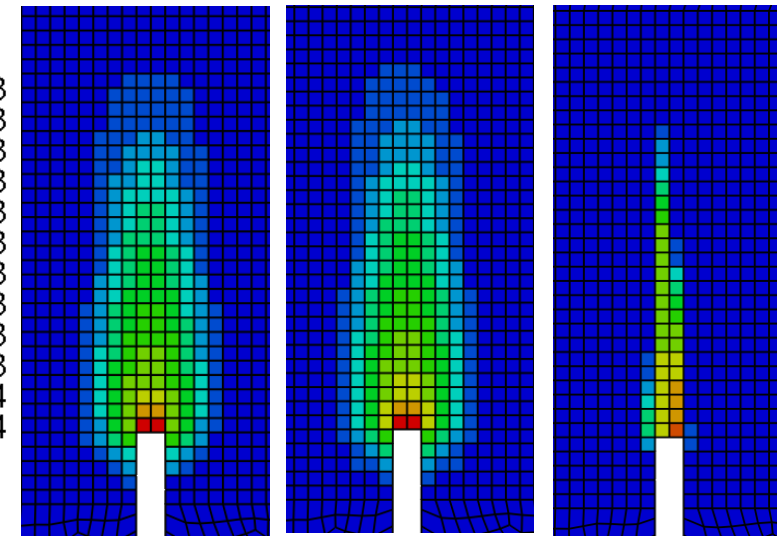


3D Extension and Threshold

$x - y$ plane
curvature
control only



Sprain
threshold
parameter

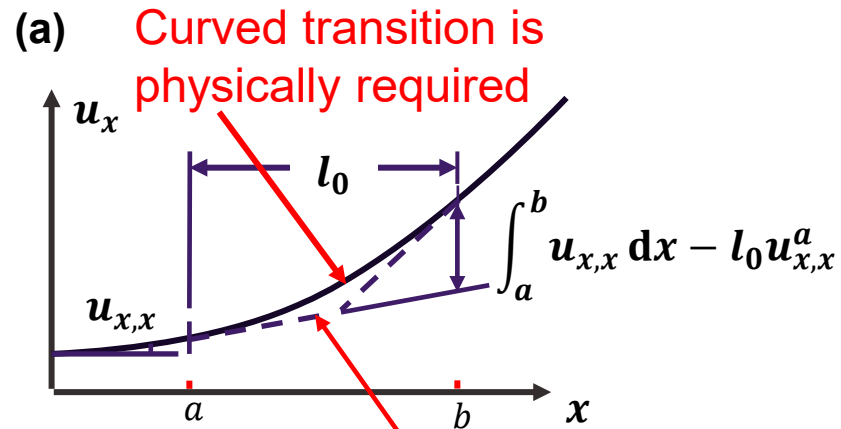


$\gamma_0 = 10^{-6}$
small

$\gamma_0 = 10^{-5}$
medium

$\gamma_0 = 1.5 \times 10^{-3}$
big

Strain-gradient vs. sICBM

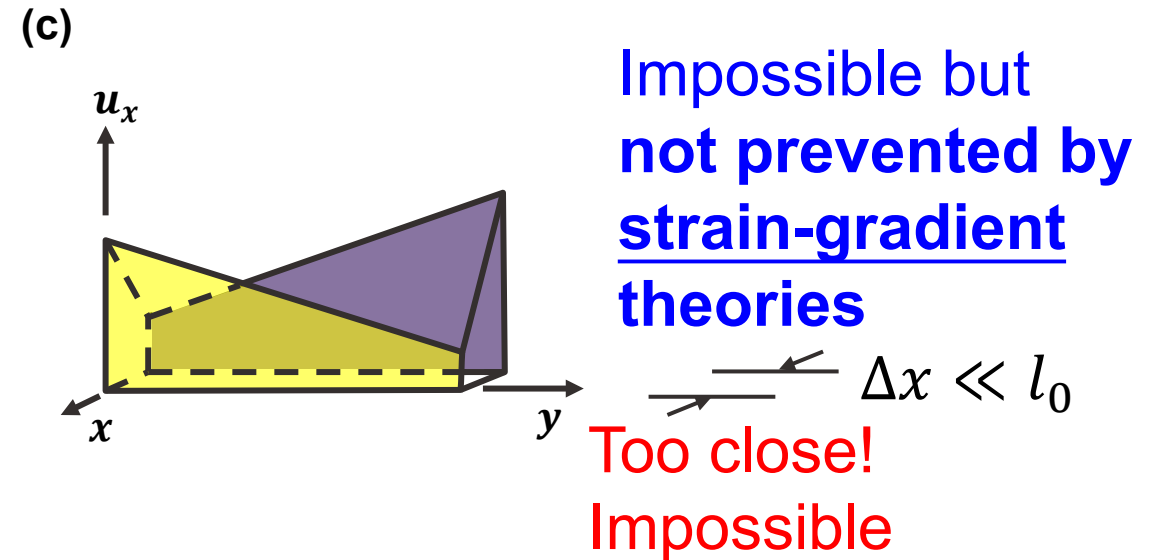
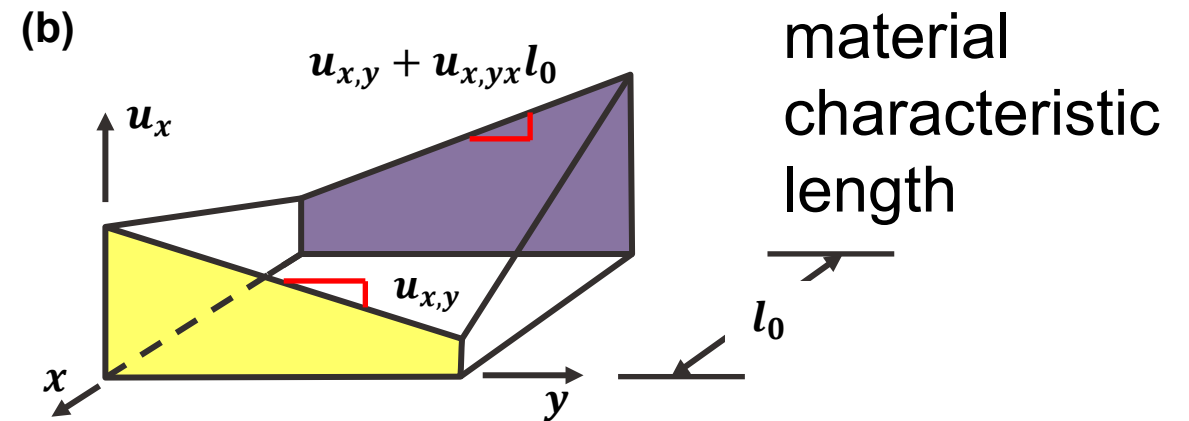


Dashed lines, with a sharp slope change, are unphysical

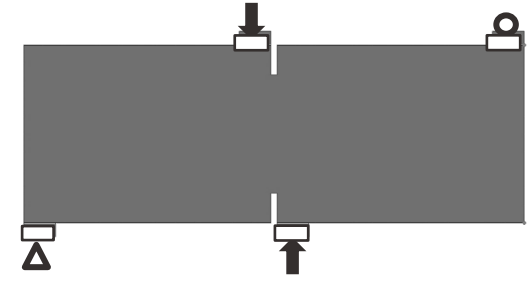
Error of strain gradient theories:

$$l_0 \epsilon_{xy,x} = l_0 u_{x,yx} - \frac{l_0}{2} (u_{x,yx} - u_{y,xx})$$

$\Delta x \rightarrow 0$ gives free torsional sliding
which must be resisted by $\omega_{xy,x}$

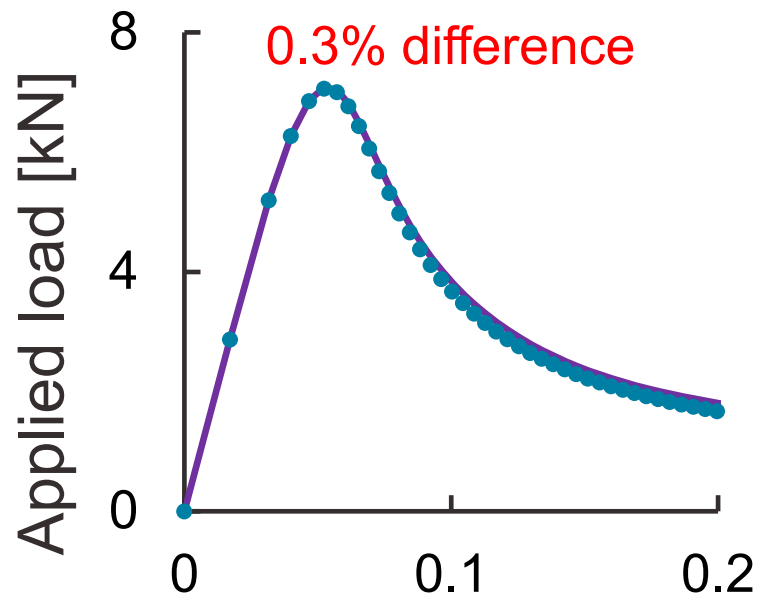


Strain-gradient vs. sICBM



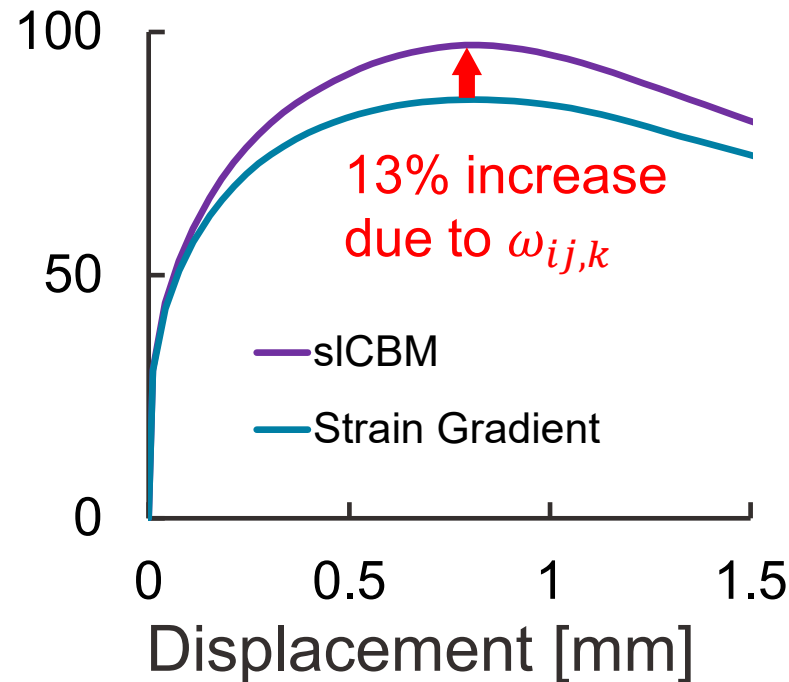
a) 3PB test

Crack line symmetry
enforces $\omega_{ij,k} = 0$

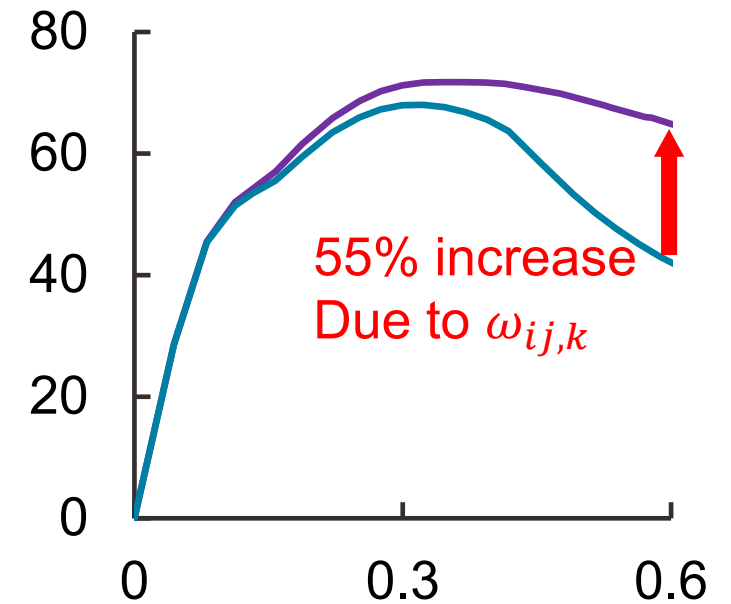


b) Shear test 1

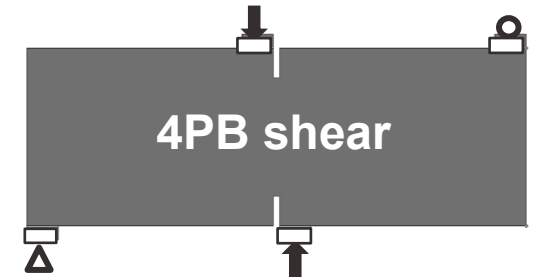
No symmetry



c) Shear test 2

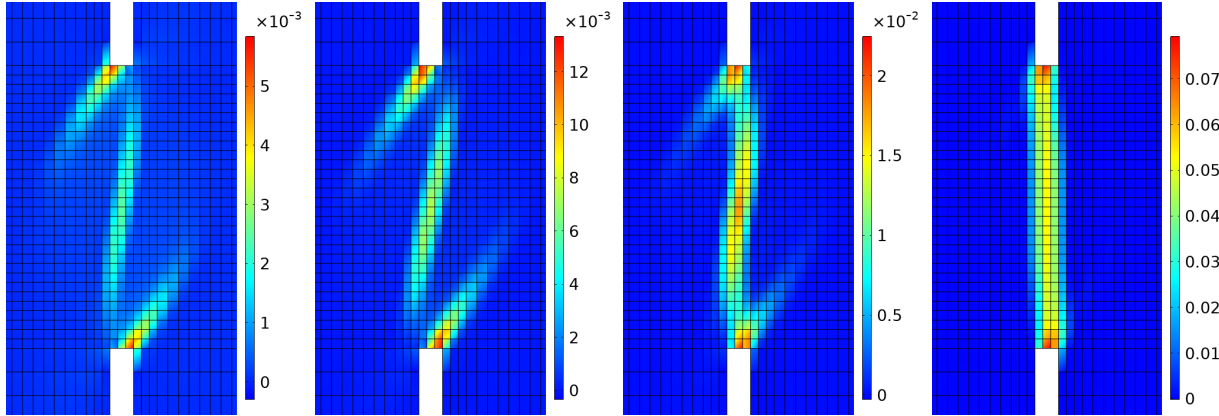


Fresh Result: Strain-gradient vs. sICBM

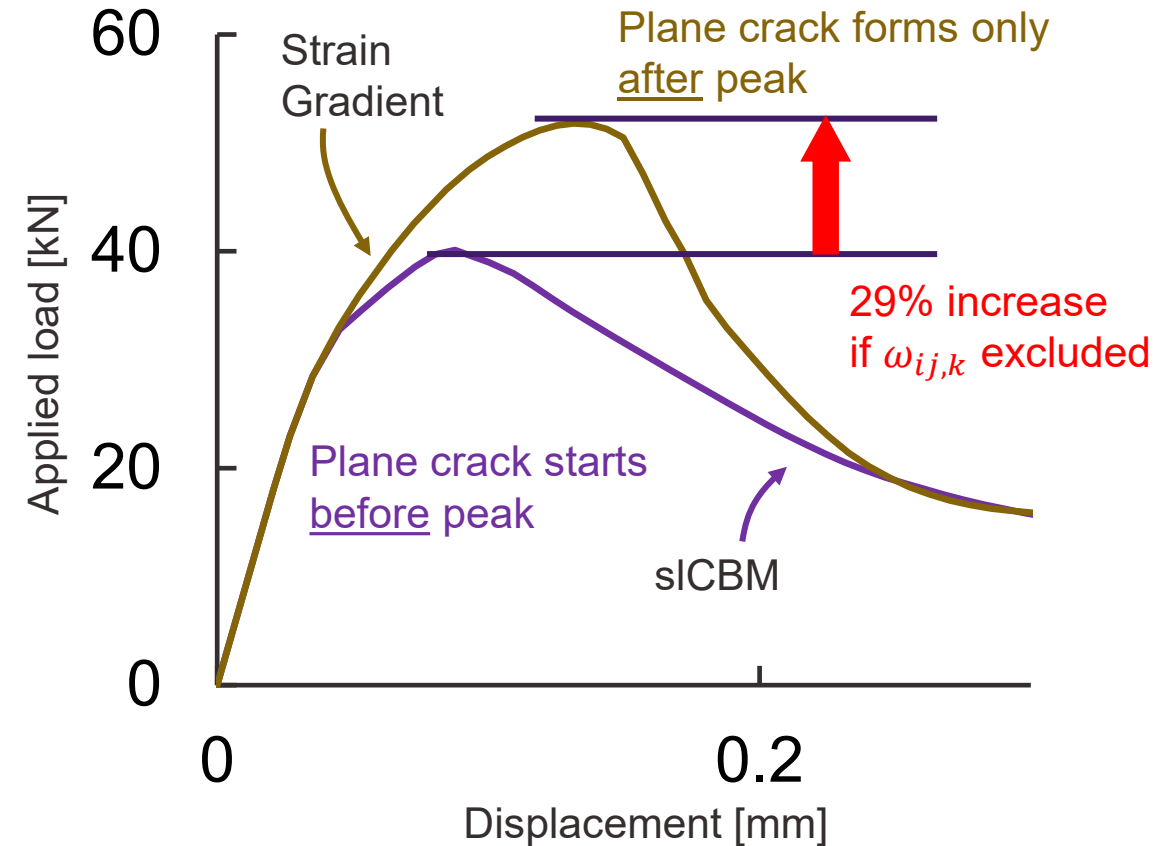
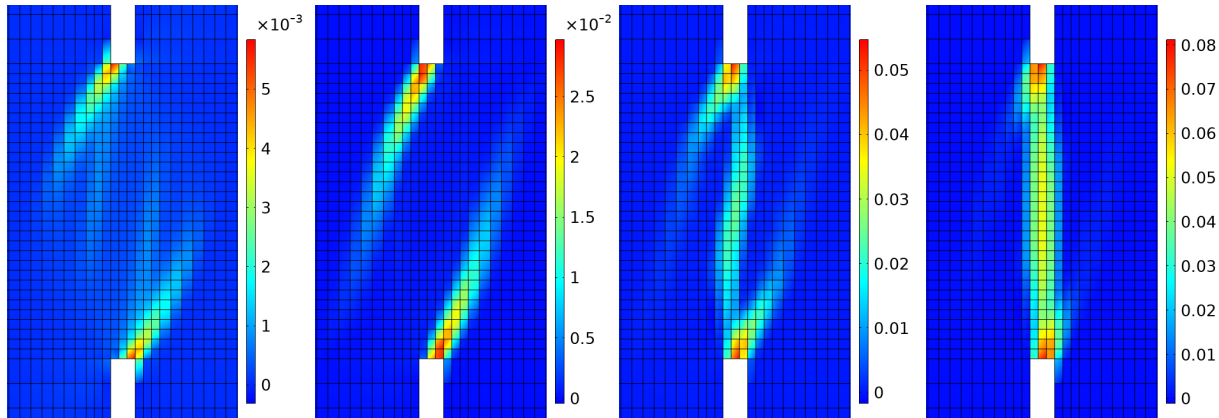


Before peak At peak load Post peak Final pattern

sICBM (sprain)



Strain Gradient (crack pattern incorrect)



Conclusions

- slCBM can simulate the variation of the crack band width.
- Mesh orientation independence was demonstrated with much high efficiency.
- M7 with Abaqus UEL can be integrated.
- Gap test → finite FPZ of variable width. The line crack models, apply only for negligible crack-parallel stresses, $\sigma_{xx}, \sigma_{yy}, \sigma_{zz} \approx 0$
- The strain-gradient models may overestimate the failure load in nonsymmetric quasibrittle failures with shear-compression by up to 55%. For symmetric failures, though, the difference is negligible.
- Constraining the independent and displacement-based gradients by Lagrange multiplier tensor allows using C_0



Thank you!

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